

SmartCity Platform Report

AI-Powered Civic Grievance System & Security Architecture

1. Project Vision & Mission

Modern municipal administrations handle thousands of civic complaints daily. Traditional grievance portals rely on manual ticket routing, which is slow, error-prone, and lacks transparent timeline forecasting. SmartCity was developed to solve this bottleneck by combining cutting-edge Artificial Intelligence with enterprise-grade privacy engineering. It provides a fast, secure, and fully automated civic ticketing pipeline that categorizes issues, forecasts resolution timelines, and notifies citizens automatically on every milestone.

2. Advanced Machine Learning (AI)

SmartCity bypasses static IF/ELSE structures in favor of mathematical models trained on historical municipal datasets. The pipeline features four distinct AI components:

- * Spam & Disposable Email Detection:** Uses a Logistic Regression model with a character N-gram Count Vectorizer to mathematically block temporary domain registrations on signup.
- * Resolution Time Forecasting:** Employs a Scikit-Learn RandomForestRegressor to estimate exact resolution timelines based on priority, season, and department trends.
- * Automated Issue Categorization:** Uses TF-IDF vectorization and Logistic Regression to route incoming description text to the correct division instantly.
- * Conversational Chatbot Assistant:** Uses Gemini 2.5 Flash as an interactive assistant with secure prompt-injection filters.

3. Firebase & Security Architecture

To guarantee security at scale, SmartCity decouples identity management from the core database. Firebase Authentication manages citizen login and registration securely. Upon successful verification, Firebase issues a cryptographically signed ID token to the citizen. The Python backend interceptor decodes this token securely to authorize citizen requests. Admins utilize a separate JWT-based cookie authorization scheme, establishing role-based access control (RBAC) to isolate citizen and official privileges.

4. Resend Automated Notification System

Communication transparency is key to civic trust. SmartCity integrates the Resend API to deliver instant email notifications directly to citizens. The system operates on a state-machine trigger: whenever an admin updates a complaint status (Accept, Resolve, Reject, Escalate), a background email task is immediately dispatched. During the sandbox/testing phase, emails are sent using onboarding@resend.dev. Due to sandbox safety rules, deliveries are strictly restricted to the verified account owner's email address. The system has a rate limit of 100 emails per day and 3,000 emails per month on the free tier.

5. Database & Relational Layout

The relational schema is implemented in MySQL (with a local SQLite fallback for testing). The complaints database model maintains strict relational integrity. To prevent credential leaks, database columns save citizen details in plain-text while core endpoints are secured with SlowAPI rate-limiting to guard against automated attacks and spam.

6. Live Admin Dashboard Statistics

Total Tickets: 500	Open Issues: 199	Resolved: 301	Avg Resolution: 11.3 Days
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Category Distribution

Road: 139 complaints
Waste: 101 complaints
Water: 100 complaints
Electricity: 68 complaints
Drainage: 41 complaints

Priority Distribution

Medium Priority: 152 tickets
Low Priority: 151 tickets
High Priority: 120 tickets
Critical Priority: 77 tickets

Tech Stack & Deployment Matrix

Layer	Technologies	Deployment / Hosting
Frontend (Citizen & Admin)	Next.js, Tailwind CSS, Recharts, Framer Motion	Vercel
Backend API	FastAPI (Python), SQLAlchemy, SlowAPI	Railway
Database	MySQL, SQLite fallback	Railway MySQL
Authentication & Email	Firebase Auth, JWT (Admin), Resend API	Firebase / Resend